

Section 1: Theory Questions & Answers

Green's Theorem

Q1. State Green's Theorem in the plane. **A1.** $\oint_C P dx + Q dy = \iint_R (\partial Q / \partial x - \partial P / \partial y) dA$, for a positively oriented, simple closed curve C enclosing region R and $\mathbf{F} = (P, Q)$ with continuous partial derivatives.

Q2. What conditions must the region satisfy for Green's Theorem? **A2.** Simply-connected region, bounded by positively oriented simple closed curve; P and Q have continuous partial derivatives.

Q3. Difference between circulation and flux forms. **A3.** Circulation: measures total rotation along boundary; flux: measures outward flow through boundary.

Parametrization of Surfaces

Q4. Define parametric surface. **A4.** $\mathbf{r}(u, v) = (x(u, v), y(u, v), z(u, v))$; simplifies curved surface computations.

Q5. Tangent vectors and normal vector. **A5.** Tangent vectors: $\mathbf{r}_u = \partial \mathbf{r} / \partial u$, $\mathbf{r}_v = \partial \mathbf{r} / \partial v$; Normal: $\mathbf{n} = \mathbf{r}_u \times \mathbf{r}_v$.

Surface Area

Q6. Surface area formula. **A6.** $\text{Area} = \iint_D |\mathbf{r}_u \times \mathbf{r}_v| du dv$

Q7. Why cross product? **A7.** Magnitude equals area of infinitesimal parallelogram spanned by tangent vectors.

Scalar Surface Integral

Q8. Define scalar surface integral. **A8.** $\iint_S f(x, y, z) dS$; sums scalar function over a surface (e.g., total heat).

Q9. Difference from vector surface integral. **A9.** Scalar multiplies function by area; vector measures flux along normal.

Divergence Theorem

Q10. State Divergence Theorem. **A10.** $\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_V \nabla \cdot \mathbf{F} dV$; total outward flux = sum of divergences inside volume.

Q11. Physical meaning of divergence. **A11.** Net outflow per unit volume; sources positive, sinks negative.

Stokes' Theorem

Q12. State Stokes' Theorem. **A12.** $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS$; circulation along boundary = total rotation through surface.

Q13. Relation to Green's Theorem. **A13.** Green's Theorem is 2D version when surface lies in xy-plane.

Q14. Importance of orientation. **A14.** Boundary must follow right-hand rule relative to normal; wrong orientation reverses sign.

Q15. Choosing a surface to simplify a line integral. **A15.** Pick surface bounded by the curve that makes $(\nabla \times \mathbf{F}) \cdot \mathbf{n}$ easiest to integrate.

Section 2: Theorem Applicability Questions

Q16. Vector field $\mathbf{F} = \langle y, -x \rangle$, curve C unit circle. Can Green's Theorem be applied? **A16.** Yes; field is continuous, region enclosed is simply-connected.

Q17. $\mathbf{F} = \langle y/(x^2 + y^2), -x/(x^2 + y^2) \rangle$, unit circle around origin. Green's Theorem? **A17.** No; discontinuity at origin.

Q18. $\mathbf{F} = \langle x^2, y^2, z^2 \rangle$, cube $0 \leq x, y, z \leq 1$. Divergence Theorem? **A18.** Yes; continuous partials, closed volume.

Q19. $\mathbf{F} = \langle x/(x^2 + y^2 + z^2), y/(x^2 + y^2 + z^2), z/(x^2 + y^2 + z^2) \rangle$, sphere containing origin. Divergence Theorem? **A19.** No; undefined at origin.

Q20. $\mathbf{F} = \langle -y, x, 0 \rangle$, unit circle in xy-plane. Stokes' Theorem? **A20.** Yes; surface can be unit disk in xy-plane; continuous partials.

Q21. $\mathbf{F} = \langle x/(x^2 + y^2), y/(x^2 + y^2), z \rangle$, curve encircling z-axis. Stokes' Theorem? **A21.** No; undefined on z-axis.

Q22. Explain why simply-connected regions matter. **A22.** Theorems like Green's or Stokes require no "holes" in the region for their integrals to properly relate boundary to interior/flux/rotation.

Q23. Given $\mathbf{F} = \langle y, -x, z \rangle$, can Green's, Stokes', or Divergence Theorem be applied to a closed curve in xy-plane? **A23.** Green's Theorem: yes (2D field in xy-plane, continuous), Stokes' Theorem: yes (surface can be disk), Divergence Theorem: no (curve not bounding a volume).

Q24. What are the main checks before applying these theorems? **A24.** Continuity of partial derivatives; smoothness of surface or curve; simply-connectedness or closed boundary; defined domain of vector field.

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